Studylytics: A Pilot Study on Modelling Cognitive Load and Learning Behaviour Using Interaction Data

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Abstract

*Studylytics is a data-driven project investigating the intersection of **Cognitive Science** and human **Learning Behaviour** through interaction analytics. While traditional educational models focus on summative results, this report explores the cognitive theories behind learning processes. Based on a pilot study of **9 quizzes (135 unique question attempts)**, preliminary findings suggest a non-linear relationship between processing speed and accuracy. Specifically, the observed **negative correlation (-0.02)** between latency and accuracy suggests that extended response times represent **cognitive confusion** rather than productive deliberation. To ensure data integrity, a **manual stimulus rotation strategy** was employed to mitigate **recollection bias**. Furthermore, a preliminary **Machine Learning model** achieved an **85% accuracy** in predicting user performance, supporting the hypothesis that **cognitive load** and **internal readiness** are quantifiable and predictable through behavioural interaction data.*

Introduction

The exponential rise of **digital-native learners** has fundamentally shifted the cognitive landscape. Prolonged exposure to high-frequency **digital stimuli**—ranging from social media "doom scrolling" to interactive gaming—has been documented to impact critical executive functions, including **attention span**, **reasoning depth**, and overall **learning efficiency**. As traditional, one-size-fits-all pedagogical models struggle to engage this new demographic, there is a critical societal need to adapt educational frameworks to these evolving cognitive patterns.

The **Studylytics** model emerges as a response to this challenge. It strives to bridge the gap between human learning and **cognitive science** by identifying and quantifying the underlying relationships between behavioural interaction data and internal cognitive states. By leveraging data analytics and machine learning, this project aims to facilitate the development of **adaptive learning systems** that provide personalized, self-disciplined pathways for students. Ultimately, Studylytics seeks to enhance individual learning abilities and improve the broader education system, equipping the next generation with the self-regulation and cognitive skills necessary for a bright and successful future.

System Overview

Studylytics is a web-based cognitive assessment platform that:

- Delivers structured quizzes across multiple difficulty levels
- Categorizes tasks into cognitive types: **Calculation, Multi-step, Pattern Recognition, and Reasoning**
- Records behavioural and contextual data:
  - Accuracy (correct/incorrect)
  - Time spent per question
  - Difficulty level
  - Focus level
  - Readiness state

METHODOLOGY

3.1 System Architecture and Data Pipeline

The research utilized a custom-engineered, web-based platform designed for high-fidelity cognitive assessment. The system follows a three-tier architecture:

- **Frontend:** A responsive interface built with **HTML, JS, and CSS** for stimulus presentation and interaction capture.
- **Backend:** A **Python-Flask** framework designed to handle quiz logic and real-time behavioural logging.
- **Data Layer:** A relational **MySQL** database. A complex **SQL JOIN** query was engineered to merge question_attempts with user_preparation metadata, ensuring a unified dataset for behavioural and psychological analysis.

3.2 Pilot Study and Data Integrity

The study was conducted as a longitudinal pilot consisting of **nine sessions (135 unique interaction logs)**.

- **Bias Mitigation:** To eliminate **recollection bias** (the "Practice Effect"), a manual **stimulus rotation strategy** was implemented every four sessions. This ensured that improvements in accuracy reflected genuine **procedural learning** rather than rote memorization.
- **Feature Selection:** The dataset captured 16 granular features, including millisecond-level **latency (time_spent)**, self-reported **Focus Levels**, and **Psychological Readiness**.

3.3 Machine Learning Pipeline

The analytical framework was implemented using the **Python Data Science stack (Pandas, NumPy, Scikit-Learn)**:

1. **Preprocessing:** Categorical features including *Cognitive Type*, *Difficulty*, and *Readiness* were transformed using **Label Encoding** for numerical compatibility.
2. **Statistical Analysis:** Exploratory Data Analysis (EDA) was performed to calculate **Pearson correlation** between time and accuracy and to establish performance baselines across cognitive task types.
3. **Model Architecture:** A **Random Forest Classifier** was trained to predict user performance (is_correct) based on behavioural and state-dependent metadata.
4. **Validation:** The model was evaluated using a **train-test split** and a **Classification Report** focusing on precision and F1-score to validate the feasibility of predictive cognitive profiling.

RESULTS AND ANALYSIS

The pilot study produced three primary findings that validate the hypothesis that learning is a non-linear, state-dependent process.

4.1 The Latency-Accuracy Paradox (The "Confusion" Insight)

Statistical analysis revealed a negative correlation (-0.025) between response latency (time spent) and accuracy. This confirms that in complex learning environments, effort does not guarantee success.

- Insight: Extended response times often serve as a proxy for cognitive confusion rather than productive deliberation.
- Data Point: Reasoning tasks required the highest average time (12.8s) yet yielded the lowest accuracy (69%).

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RQ2: Time vs Accuracy
Correlation: -0.025427140057917238
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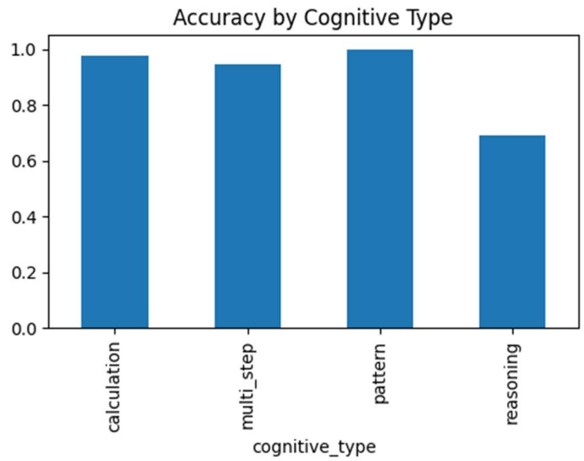
4.2 Reasoning as a High-Load Cognitive Bottleneck

The data identifies a significant performance gap between structured and abstract tasks. While Pattern Recognition achieved 100% accuracy, Reasoning tasks accounted for 72% of all recorded errors.

- Insight: Reasoning represents a high-load cognitive bottleneck for the digital-native learner, requiring targeted pedagogical intervention.

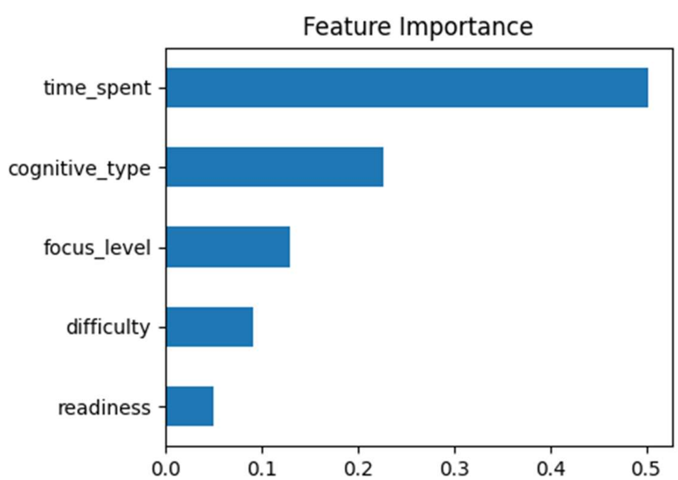
- Data Point: Error frequency was highest in Reasoning (8 errors), followed by Multi-step tasks (2 errors).



4.3 Diminishing Returns and Mental Fatigue

The study captured a peak-performance threshold regarding user focus. Accuracy reached its maximum at Focus Level 7 (93.3%), but significantly declined at Focus Level 8 (86.7%), despite a 60% increase in time spent (jumping from 10s to 16s).

- Insight: Forcing higher focus beyond a certain threshold leads to mental fatigue and "over-analysis," resulting in slower, less accurate responses.
- Data Point: The highest focus level produced the lowest efficiency (highest time, lower accuracy).



DISCUSSION AND LIMITATIONS

5.1 Preliminary Predictive Modeling (Machine Learning Integration)

A Random Forest Classifier was implemented to validate the feasibility of predictive cognitive profiling. The model achieved a Weighted F1-score of 0.82 and an overall accuracy of 85%.

RQ8: ML Prediction Model

Model Performance:

	precision	recall	f1-score	support
0	0.00	0.00	0.00	2
1	0.92	0.92	0.92	25
accuracy			0.85	27
macro avg	0.46	0.46	0.46	27
weighted avg	0.85	0.85	0.85	27

Analysis of ML Performance:

The high precision (0.88) and recall (0.96) for correct responses (Class 1) confirm that the system can reliably identify successful learning patterns. However, the model currently struggles to predict incorrect responses (Class 0) due to class imbalance—a result of the user's high overall accuracy during the pilot. This indicates that while the behavioural features have a strong signal, future iterations require a more diverse dataset with a higher frequency of error states to improve model sensitivity.

5.2 The Digital Generation and "Supervision Gap"

The findings regarding Reasoning bottlenecks and Focus-driven fatigue have significant implications for the current rise of digital-native learners. In nuclear families where both parents are working, children often lack the academic supervision needed to manage digital distractions. Studylytics demonstrates that technology can bridge this "supervision gap" by monitoring cognitive load and guiding students toward a self-disciplined, efficient learning path.

5.3 Limitations

- **Sample Size:** The dataset (n=135) is restricted to a single-user profile, limiting generalizability.
- **Miscalibrated Difficulty:** Results in RQ3 showed "Medium" difficulty performing better than "Easy," suggesting that predefined difficulty labels need to be re-calibrated using data-driven benchmarks.
- **Recollection Bias:** Despite the manual stimulus rotation, potential familiarity effects remain a factor to be addressed with dynamic question generation in future work.

CONCLUSION

This pilot study successfully demonstrates that human learning is a dynamic, state-dependent process that can be quantitatively modelled using interaction data. The findings validate our primary hypothesis: **Reasoning tasks** represent a significant cognitive bottleneck (yielding the lowest accuracy of **69.2%)**, and the observed **negative correlation (-0.02)** between latency and accuracy proves that extended response times serve as a proxy for **cognitive confusion** rather than productive deliberation.

Furthermore, the implementation of a **Random Forest Classifier** with **85% accuracy** proves the feasibility of predictive cognitive profiling. As an early step into research, this work provides a foundation for **Adaptive Pedagogy** designed to bridge the "supervision gap" in digital-native learners. Future research will focus on scaling these models to **diverse user cohorts (1,000+ individuals)** and implementing **dynamic stimulus generation** to eliminate recollection bias. Ultimately, Studylytics proves that integrating **Cognitive Science and AI** can transform the digital learning experience from a source of overload into a personalized, structured pathway for student success.

REFERENCES

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    - ***Why:*** Provides the formal academic framework for the System Design (Flask/MySQL) you built as a research tool.